

Mason Bane

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Summary: Aspiring defense industry engineer with a strong foundation in computer engineering, embedded systems, and hardware/software integration. Passionate about serving the United States by advancing cutting-edge technologies in embedded software, simulation, and performance-critical computing to strengthen national defense capabilities. U.S. Citizen and eligible for DoD clearance.

EDUCATION

Tarleton State University

Stephenville, TX

Bachelor of Science in Computer Science, Concentration in Computer Engineering Aug. 2022 – May 2026 (Expected)

Minor in Mathematics

GPA: 3.94/4.00 (Institutional) — Cumulative: 3.75/4.00

Hill College

Hillsboro, TX

Associate of Arts in Liberal Arts

Aug. 2019 – Sept. 2023

EXPERIENCE

Undergraduate Research Assistant - Lead Programmer

May 2024 – Present

Tarleton State University

Stephenville, TX

- Developed and optimized various N-Body simulations utilizing C/C++, enhancing computational efficiency and accuracy through advanced algorithms and parallel processing with CUDA.
- Collaborated with interdisciplinary teams to create digital twins to model complex problems using OpenGL and Blender, translating scientific concepts into interactive simulations and improving data interpretation.
- Conducted rigorous testing and debugging of simulation software, ensuring robust performance and reliability while documenting processes to facilitate knowledge transfer and future research initiatives.

Undergraduate Technology Specialist - HPC Lab Manager

Aug. 2024 – Present

Tarleton State University

Stephenville, TX

- Managed and maintained 15 Linux devices in a high-performance computer lab dedicated to research initiatives
- Performed system updates, hardware maintenance, and technical support, resolving issues promptly to minimize downtime and maintain optimal performance.
- Maintained a clean and organized lab environment, promoting a collaborative workspace that fosters innovation and productivity.

PROJECTS

N-body Digital Twin of the Left Atrium | NIH Grant #1R15HL179671-01 | C, CUDA Aug. 2024 – Present

- Engineered a parallel N-body model of the left atrium using CUDA, with a 20,000+ node mesh to simulate atrial arrhythmias in near real-time.
- Developed an intuitive C++/ImGui interface to control simulation parameters and visualize outputs, bridging the gap between complex computational models and end-user (research/clinical) requirements.
- Presented findings at academic conferences, demonstrating the tool's potential to improve clinical decision-making and transform training for medical professionals.

Secure IoT Sensor Node with Hardware TrustZone | C, STM32CubeIDE/HAL, I2C/SPI Dec. 2025 – Present

- Implementing a secure data pipeline on an STM32H5 MCU, utilizing Arm TrustZone and the Secure Manager to create a hardware-rooted chain of trust for environmental sensor data.
- Developing drivers and application logic to interface with a Bosch BME280 temperature, humidity, and pressure sensor via I2C communication protocol.
- Structuring firmware to cryptographically sign sensor readings within the secure processing environment for tamper-evident logging.
- Utilizing STM32CubeIDE and the HAL for peripheral configuration and hardware security module (HSM) management as a foundation for secure telemetry systems.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, ARM Assembly (Cortex-M), Java, Bash

Microcontroller Platforms: STM32 (H5/Cortex-M33), TIVA-C (TM4C), Arduino, Raspberry Pi

Embedded Protocols & Tools: I2C, SPI, UART, GPIO, STM32CubeIDE, GDB, RTOS Concepts

Systems & Development: Git/GitHub, Linux, CMake/Make, Visual Studio, VS Code

Simulation & Design: LTSpice, MATLAB, CUDA, OpenGL, ImGui, Blender